

Lesson 6: Model Assessment and Deployment



6.0 Lesson Overview



Lesson Overview

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6.1 Model Assessment and Comparison



Essential Deployment Tasks



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Essential Deployment Tasks



Essential Deployment Tasks

- Assess models.
- Compare models.
- Score the champion model.
- Assess model bias.
- Monitor model performance over time.
- Update the model as needed.

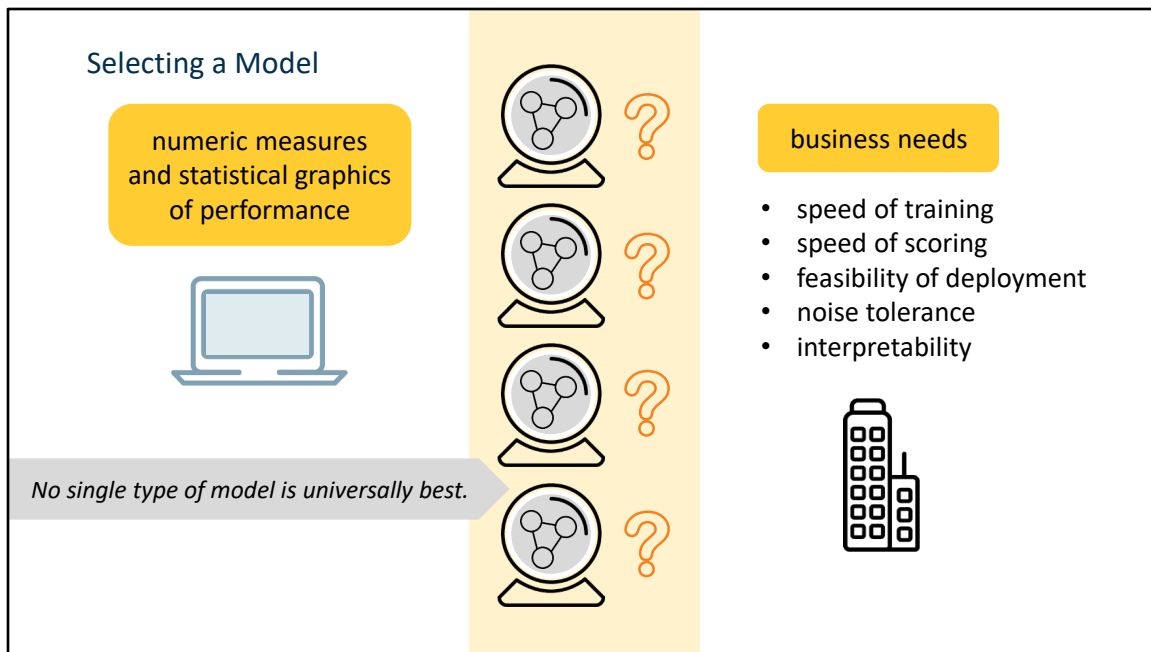
We are now at the Deployment phase of the analytics life cycle. The essential tasks for deployment start with assessing the individual models. Next, we compare the candidate models and select the one that best represents the association between the inputs and the target. This model is typically called the champion model.

Later in this lesson, you learn how to put the model into production (a process called scoring). Then you learn about assessing model bias, monitoring the model's performance over time and updating the model as needed.

Selecting a Model



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When selecting a champion model, it's important to remember that no single type of model is always the best (that is, in all situations and over time).

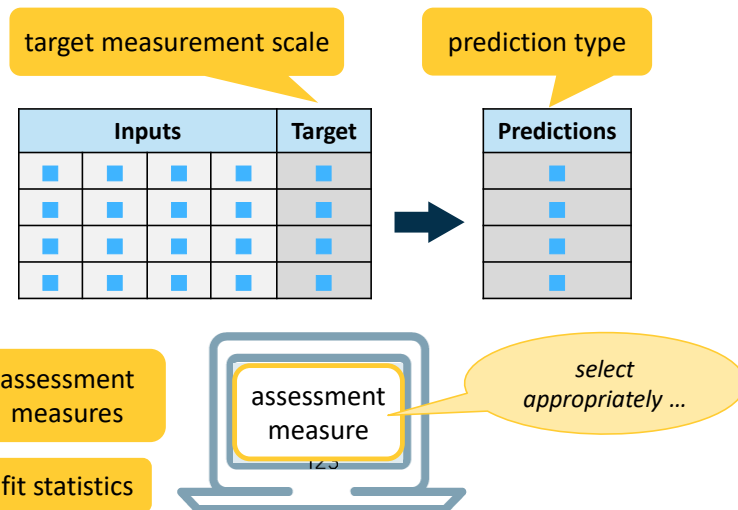
You select a model primarily based on numeric measures and statistical graphics of performance. However, you must also consider the needs of the business situation. Common business factors include the speed of training the model (for example, if the model will often need to be rebuilt), the speed of scoring, the feasibility of deployment, the noise tolerance (that is, the strength of the signal in the data), and interpretability (that is, how easy it is to explain the model's predictions).

Numeric Measures of Model Performance



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Numeric Measures of Model Performance by Prediction Type



Numeric measures used to evaluate model performance are called assessment measures or fit statistics.

To compare models, you select an appropriate assessment measure based on two factors: the target measurement scale and the prediction type.

An assessment measure that is appropriate for a binary target might not make sense for an interval target. Similarly, an assessment measure that is suitable for decision predictions cannot be used for estimate predictions.

Binary Target

Prediction Type	Assessment Measures
Decision	• Accuracy • Misclassification • Profit/loss
Ranking	• C-statistic • Gini coefficient
Estimate	• Average square error • Root mean square error • Schwarz criterion (SC) • Akaike's information criterion (AIC)

Shown here are some common assessment measures for a binary target, given different prediction types. For decision predictions, you might choose from accuracy, misclassification, or profit or loss. For rankings, you might choose from the C-statistic (which is the area under the ROC curve) or the Gini coefficient. And for estimates, you might choose from the average square error, root mean square error, Schwarz criterion, or Akaike's information criterion.

You learn more about assessment measures later.

Numeric Measures of Model Performance by Prediction Type

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Confusion Matrix for Decision Predictions



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Confusion Matrix for Decision Predictions

		Predicted Outcome		
		Primary	Secondary	
Actual Outcome	Primary	TP	FN	TP + FN
	Secondary	FP	TN	FP + TN

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

$$\text{Specificity} = \frac{TN}{TN + FP}$$

The confusion matrix is a common format for calculating and displaying assessment measures of model performance for decision predictions on a binary target. This matrix is a crosstabulation of the actual and predicted outcomes, based on a decision rule. A simple decision rule allocates cases to the target class with the greatest posterior (that is, predicted) probability. For binary targets, this corresponds to a 50% cutoff on the posterior probability.

A confusion matrix displays four counts: true positives, true negatives, false positives, and false negatives. A true positive (or TP) is a case known to be a primary outcome and also predicted as a primary outcome. A true negative (or TN) is a known secondary case predicted as a secondary case. A false positive (or FP) is a case that is predicted as a primary outcome but is actually a known secondary outcome. And a false negative (or FN) case is a case that is predicted as a secondary outcome but known to be a primary outcome.

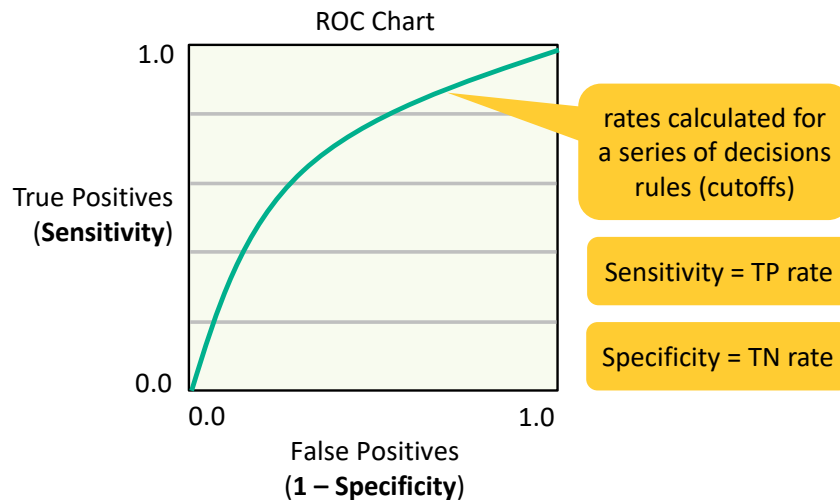
True and false positives and true and false negatives are used to calculate various assessment measures, such as sensitivity and specificity. Sensitivity, the true positive rate, is the number of true positive decisions divided by the total number of known primary cases. Specificity, the true negative rate, is the number of true negative decisions divided by the total number of known secondary cases. These measures are the basis for the ROC chart, which you learn about next.

ROC Charts and the C-Statistic



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ROC Charts and the C-Statistic

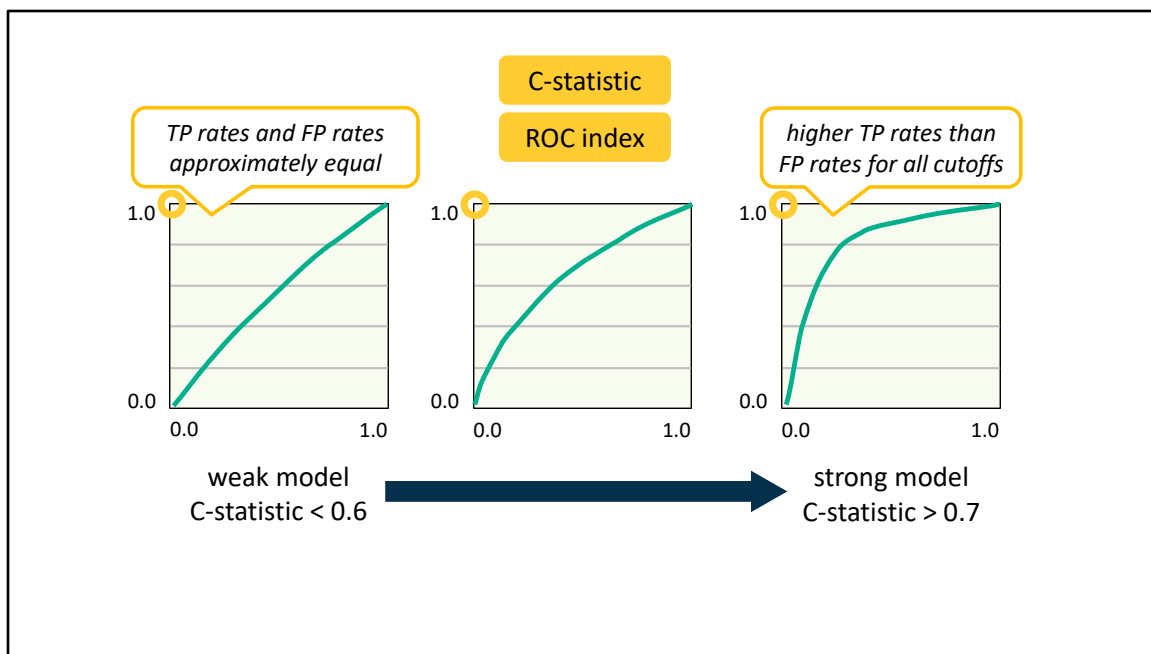


The ROC chart is a commonly used graphical representation of model performance for a binary target. ROC stands for receiver operating characteristic. ROC charts are based on measures of sensitivity and specificity. The sensitivity of a model is the true positive rate. The specificity of a model is the true negative rate.

In the ROC chart, the Y axis represents the cumulative rate of true positives (that is, **sensitivity**). The X axis represents the cumulative rate of false positives, which is **1 - Specificity**. The rate on each axis ranges from 0 to 1, so the plot is contained within a unit square. The rates are calculated for a series of decision rules (in other words, cutoffs) between 0% and 100%. However, the cutoffs are not shown in the chart.

The ROC chart plots the relationship between the change in the true positive rate and the change in the false positive rate. The closer the true positive rate is to 1 and the closer the false positive rate is to zero, the better the model. In other words, the higher the value of sensitivity is and the lower the value of 1-specificity is, the better the model.

What does this look like in the ROC chart?

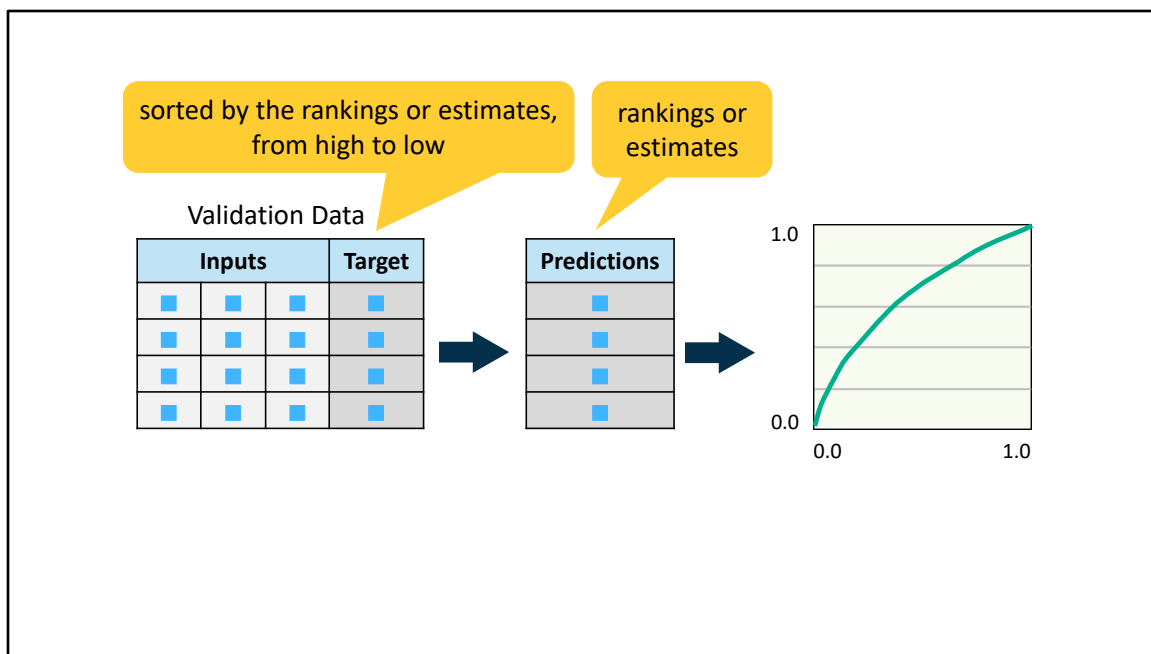


The closer the ROC curve is to the upper left corner of the chart, the better the classification accuracy of the model.

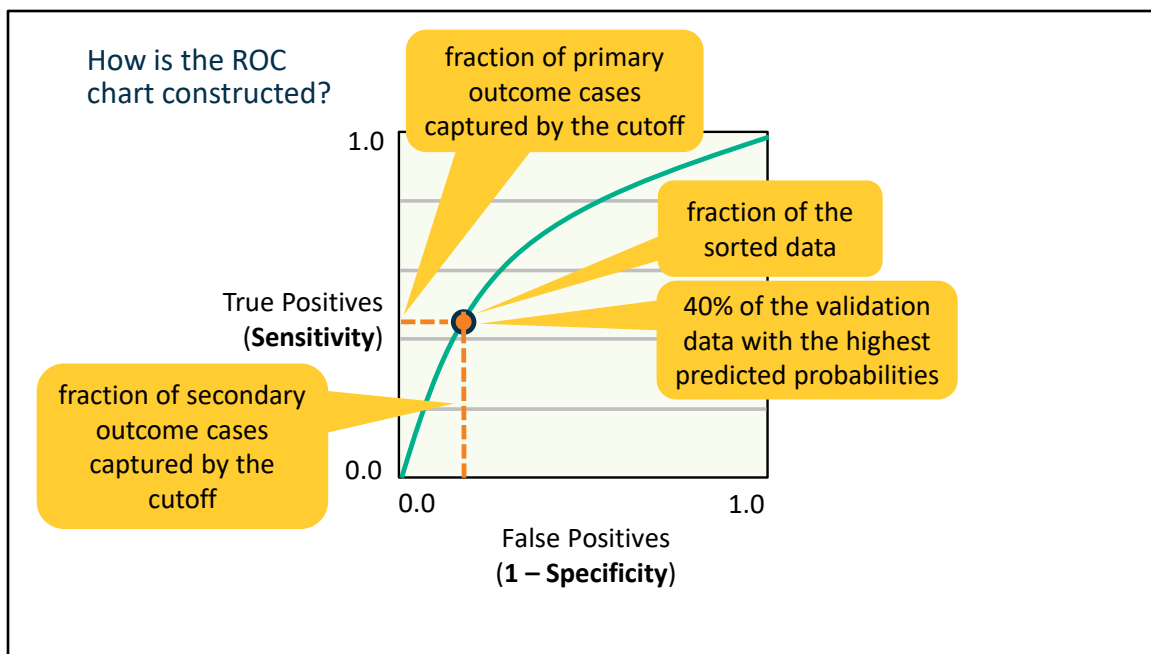
The area under the ROC curve is a measure also known as the C-statistic or the ROC index. The larger the area under the curve, the higher the C-statistic and the better the model's classification accuracy.

Models whose true positive rates and false positive rates are approximately equal are weak models. As the chart on the left shows, a weak model has a curve that is close to the diagonal between the lower left corner and the upper right corner. This weak model has a C-statistic less than 0.6.

Models that have much higher true positive rates than false positive rates for all cutoffs are strong models, as shown in the chart on the right. Notice that the curve approaches the upper left corner. This strong model has a C-statistic greater than 0.7. The area under the ROC curve summarizes the tradeoff between the primary versus secondary cases that are captured.



It is easier to interpret ROC charts when you understand how they are constructed. To create an ROC chart, predictions are generated for a set of validation data. These predictions must be rankings or estimates. Then the validation data are sorted by the ranks or estimates, from high to low. This creates a list of cases, sorted in order of importance.



Each point on the ROC chart corresponds to a specific fraction of the sorted data.

The sample point on this ROC chart corresponds to the selection of 40% of the validation data with the highest predicted probabilities. The selection value of this point cannot be seen from the chart.

The vertical, or y, coordinate of the point indicates the fraction of primary outcome cases captured by the cutoff (here, approximately 45%).

The horizontal, or x, coordinate of the point indicates the fraction of secondary outcome cases captured by the cutoff (here, approximately 25%).

The ROC chart represents the combination of similar calculations for all selection fractions.

Question: C-Statistic



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Question: C-Statistic

To use the C-statistic to assess model performance, you need to know the cutoff.

- a. True
- b. False

Answer: C-Statistic

To use the C-statistic to assess model performance, you need to know the cutoff.

- a. True
- ☒ b. False

CA: b

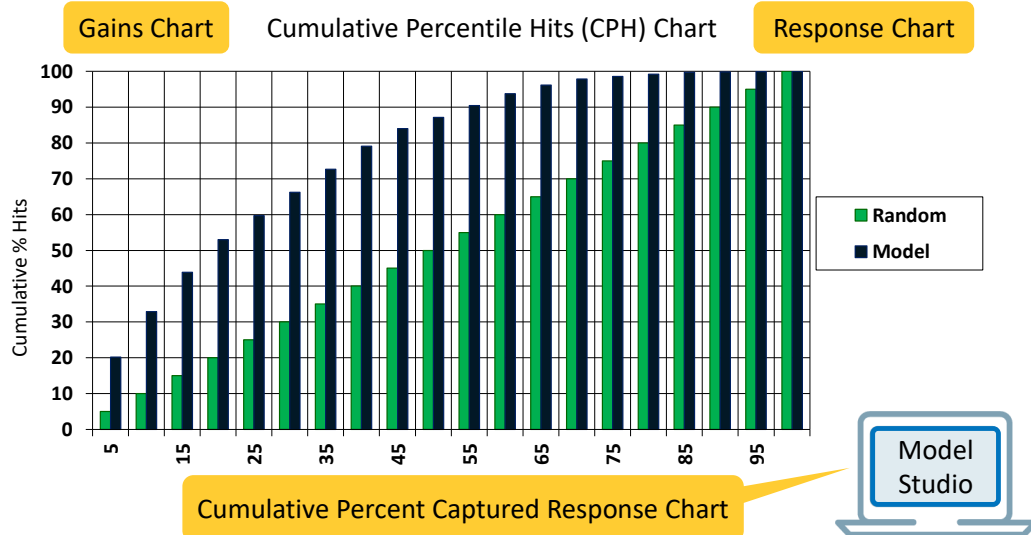
You do not need to know the cutoff to use the C-statistic. The cutoff is used in the calculation of the C-statistic.

Charts Based on Response Rate: CPH and Lift



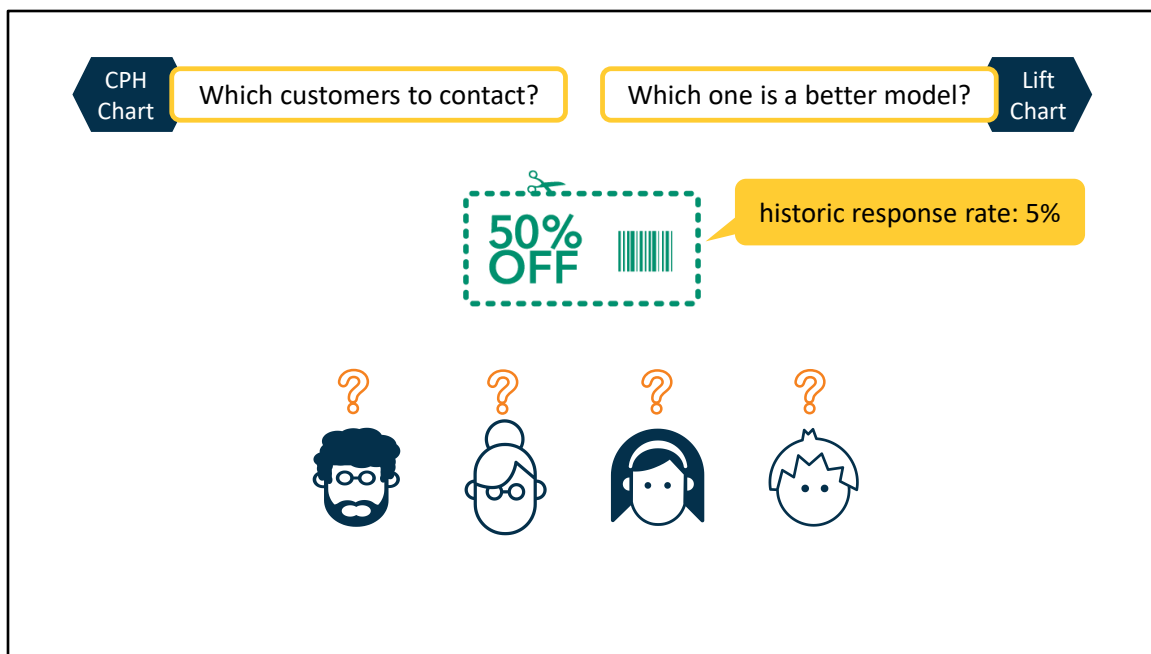
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Charts Based on Response Rate: CPH and Lift



The cumulative percentile hits chart (or CPH chart) illustrates the advantage of using a predictive model to make business decisions as compared to not using a model (that is, having a baseline or random model). The CPH chart is also known by other names, such as gains chart or response chart.

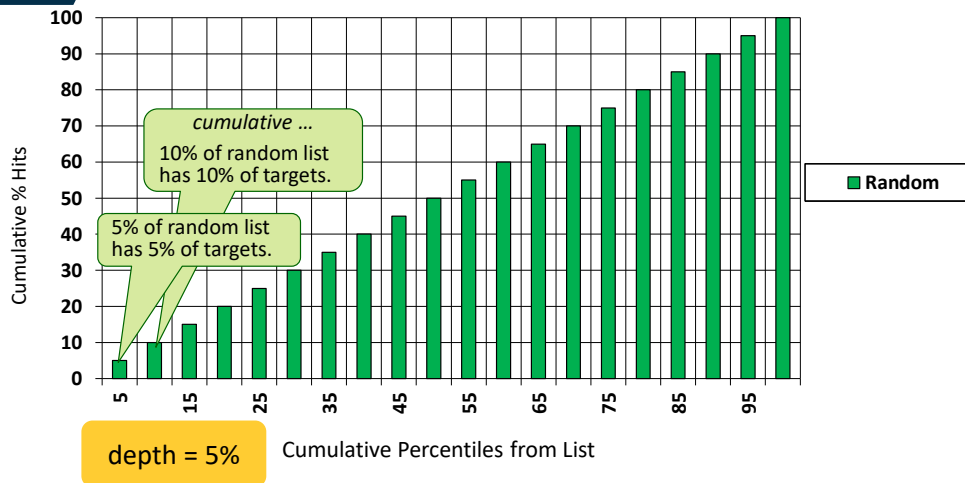
In Model Studio, this chart is called the Cumulative Percent Captured Response chart.



Suppose we want to choose which customers to contact for a direct marketing campaign. The historic response rate is 5%.

CPH
Chart

Which customers to contact?



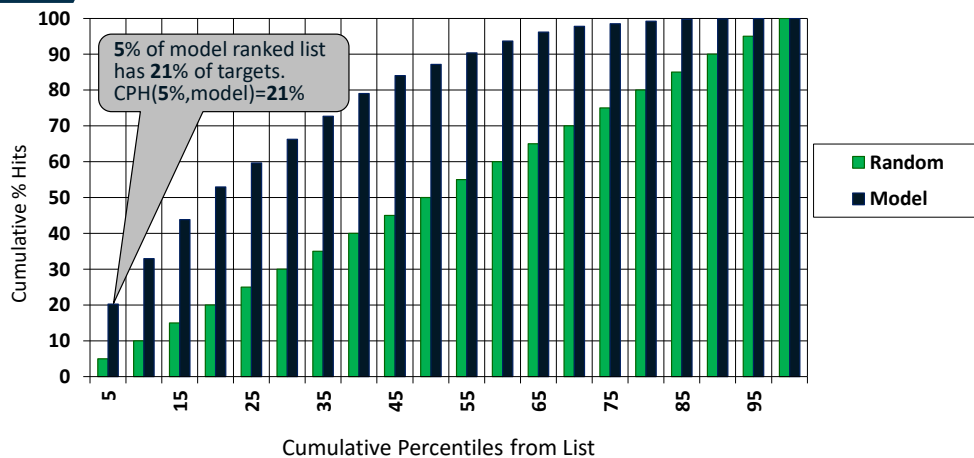
This preliminary chart shows the cumulative percentile hits when we do not use a model. The Y axis represents the cumulative percentile hits. The X axis represents the cumulative percentiles from the list of cases (in this example, customers). This particular chart has 20 percentiles (that is, 20 bars).

The first bar shows that 5% of the event cases are captured in the top 5% of customers. The selected fraction of cases (in this example, 5%) is known as the depth.

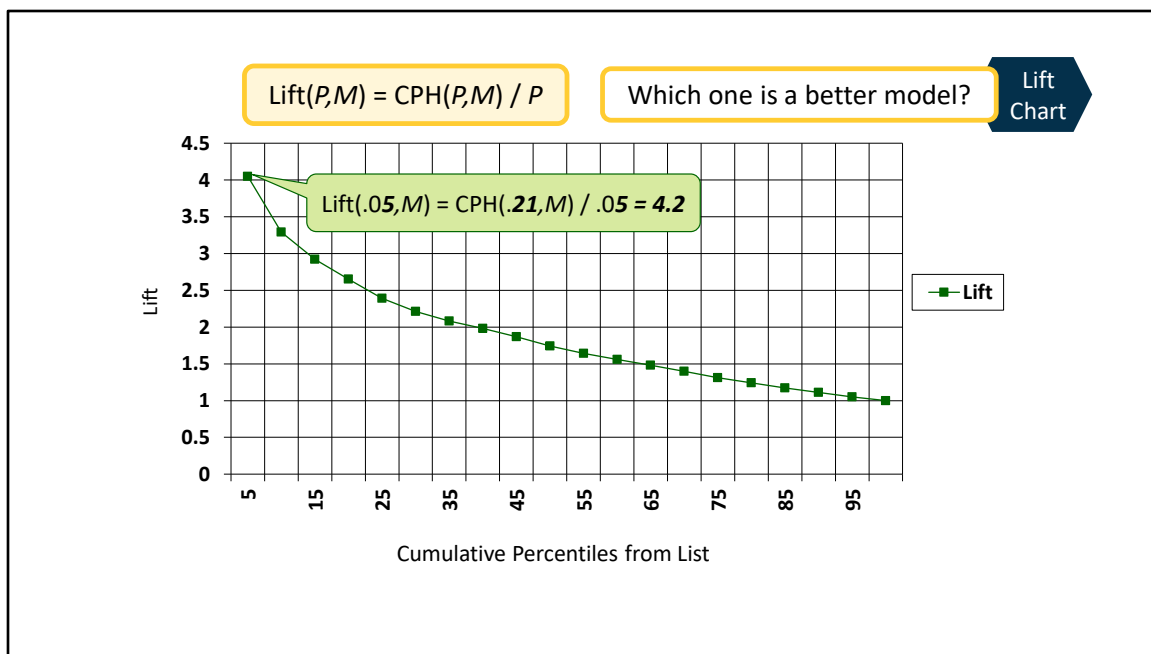
Because the graph is cumulative, the second bar shows that 10% of the event cases are captured in the top 10% of customers.

CPH Chart

Which customers to contact?



The cumulative percentile hits based on a predictive model are now added to the chart. For the top 5% of customers ranked on their posterior probabilities, the model found that 21% responded. For the top 10% of customers (again, ranked by their posterior probabilities), the model found that 32% responded, when calculated cumulatively. For all fractions of customers, the chart shows that the model captures more responders than not using a model.

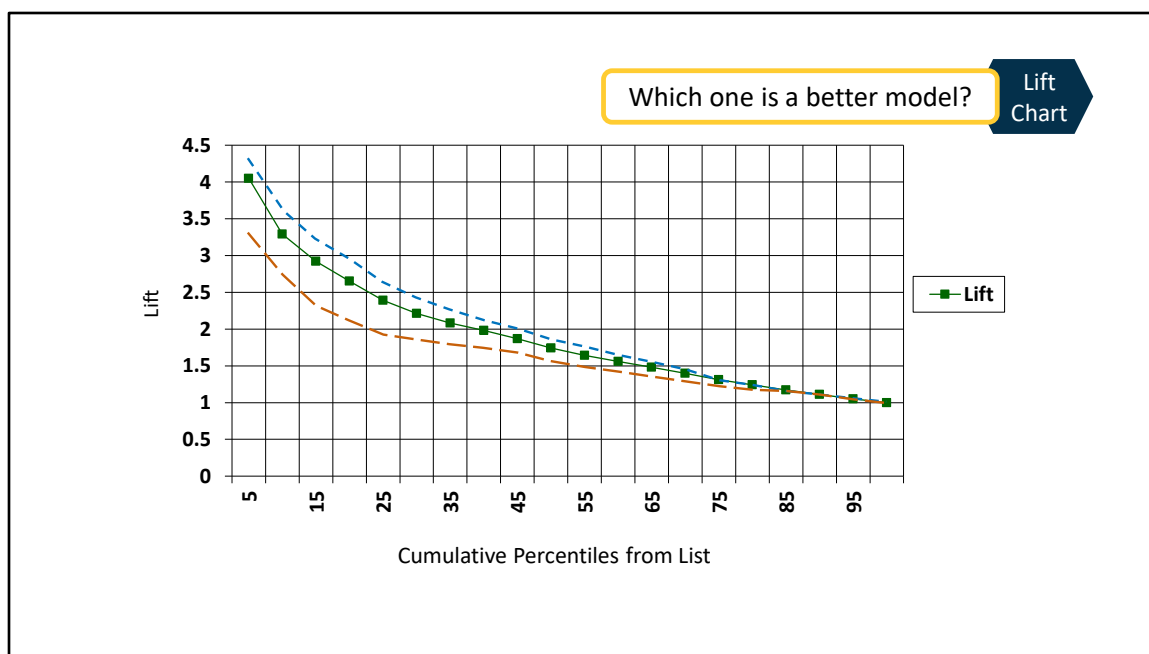


A cumulative lift chart is a graphical tool that illustrates the advantage of using a predictive model as compared to not using a model. A cumulative lift chart is often called a lift chart, as you will hear in this course. Cumulative lift is a ratio of response rates.

The response rate in the numerator equals the cumulative percentile hits for a given percentile (P) from a given model (M). The response rate in the denominator equals the cumulative percentile hits for the same percentile given no model.

In the customer response example, the CPH for the top 5% of customers, given the model, was 21%. Given no model, the response rate for the top 5% of customers (from a randomly ordered list) is 5%.

Thus, the lift at 5% is 21% divided by 5%, which is 4.2. This indicates that, for the top 5% of customers, the model captured 4.2 times as many responders, compared to not using a model.



Here is the lift chart for the customer response example. The Y axis represents the lift and the X axis represents the cumulative percentiles from the list.

A lift chart can contain lines for multiple models, which is helpful for selecting the best model in a specific business scenario. The model with the higher lift curve is generally better for model deployment. However, the best model can vary depending on the percentile. For the customer response example, if you expect to contact 20% of your customers, you want to choose the model that performs best on this percentile.

Ways of Comparing Models in Model Studio



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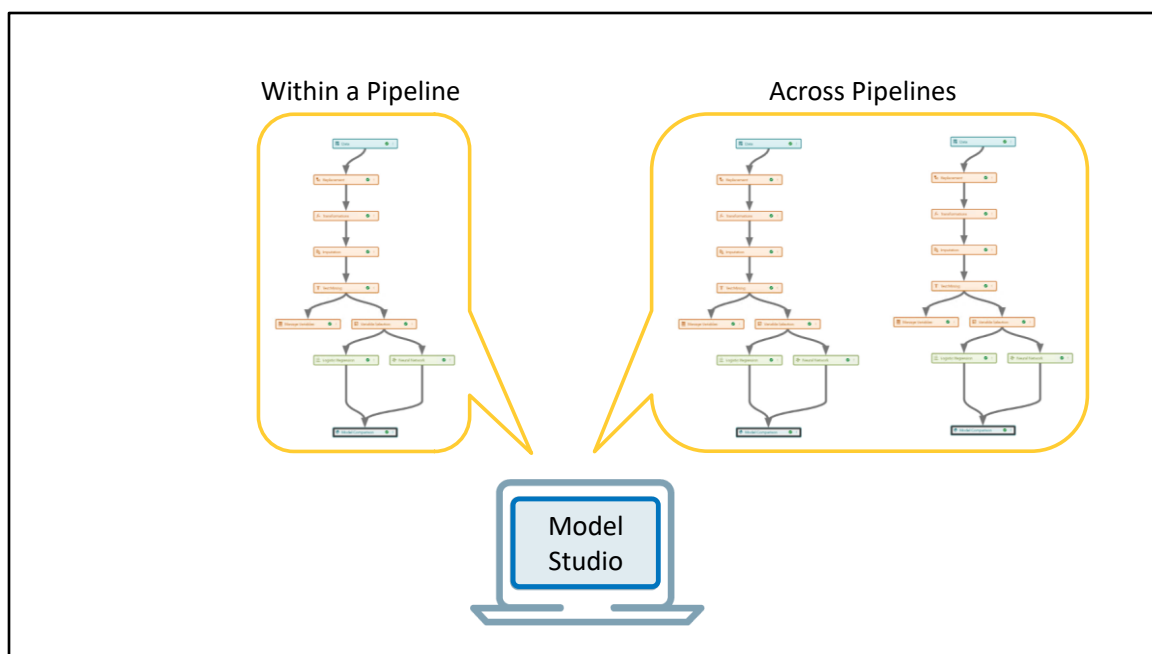
Ways of Comparing Models in Model Studio



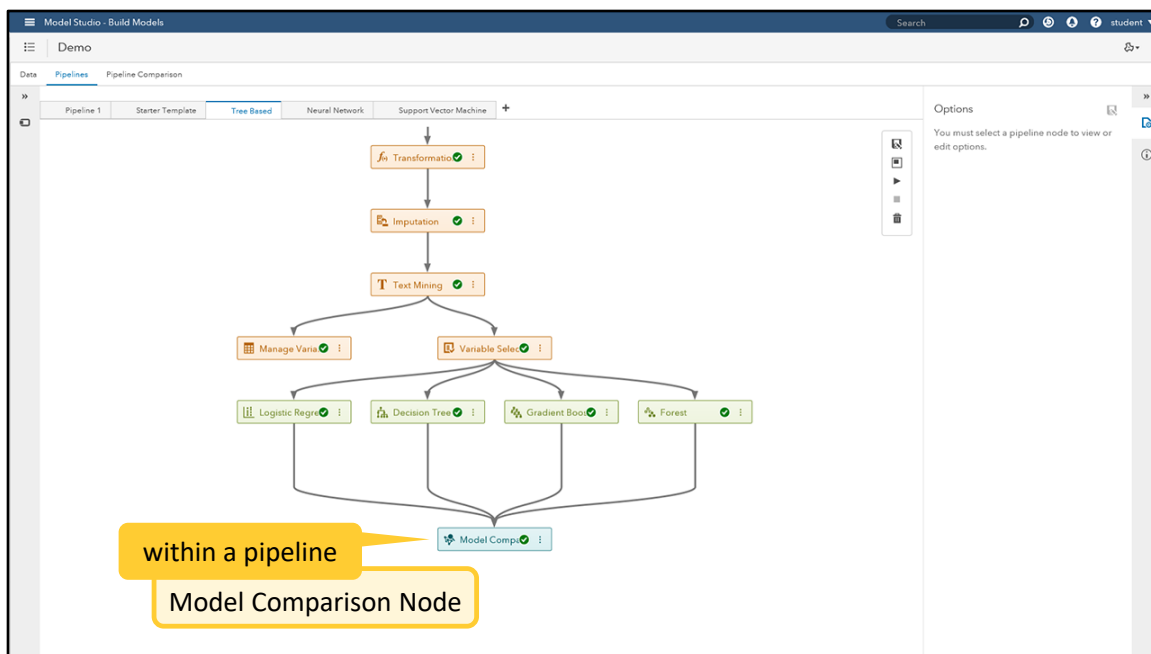
Essential Deployment Tasks

- ✓ Assess models.
- **Compare models.**
- Score the champion model.
- Assess model bias.
- Monitor model performance over time.
- Update the model as needed.

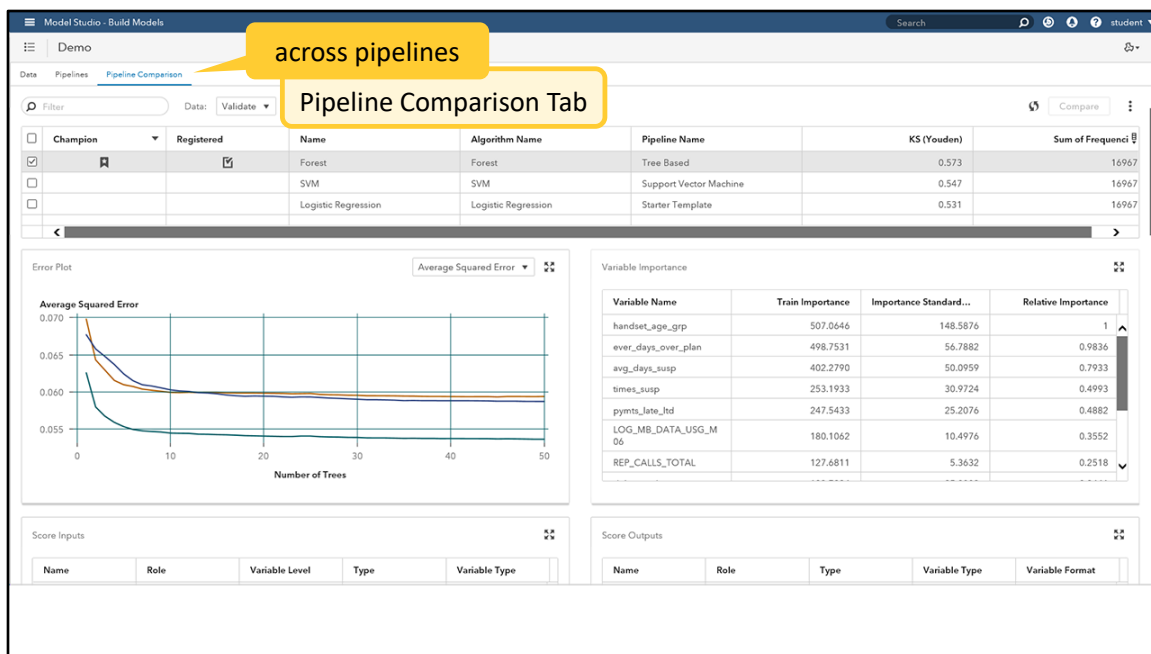
After assessing the individual models, it's time to compare them and select the champion model.



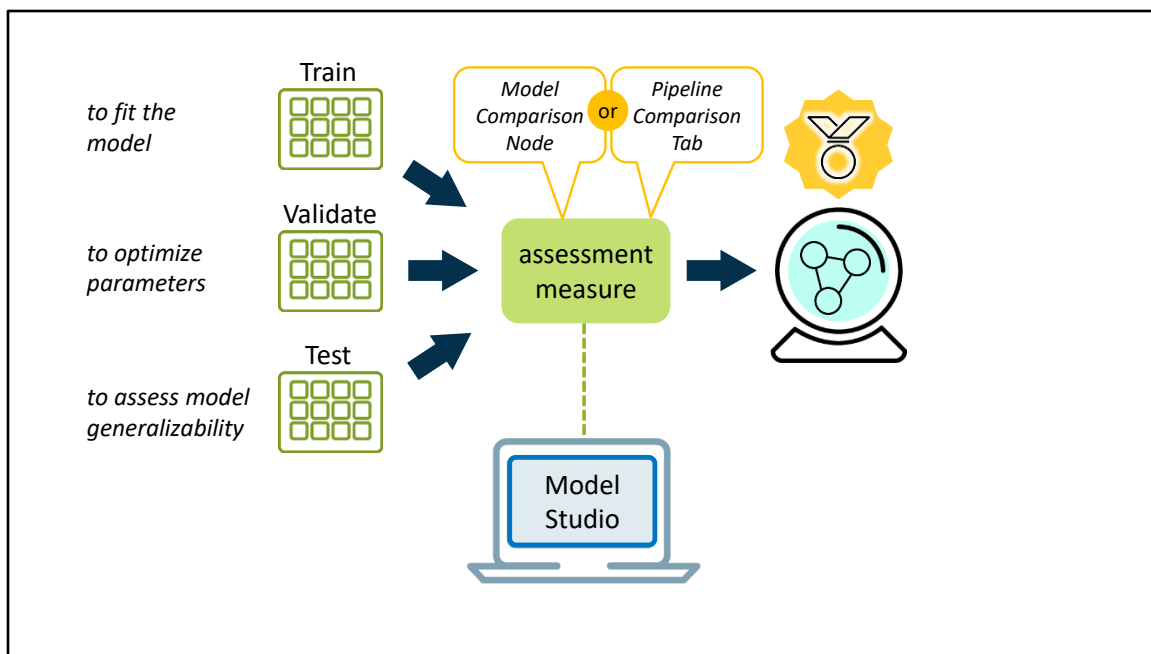
In Model Studio, you can compare models in two ways: within a pipeline and across pipelines.



To compare models within a pipeline, you use the Model Comparison node. Remember that when you add a node for any supervised learning model to a pipeline, the Model Comparison node is automatically added as well.



To compare models across pipelines, you use the Pipeline Comparison tab. By default, the Pipeline Comparison tab compares the champion models from each pipeline. However, if you want, you can add other models to the Pipeline Comparison tab.



To select a champion model either within a pipeline or between pipelines, Model Studio uses a default assessment measure, which varies by the type of target. If you want, you can specify a different assessment measure.

Model Studio computes all assessment measures for each available data partition (train, validate, and test). By default, Model Studio selects a champion model based on the validation data set unless a test data set is available. If you want, you can specify a different data set.

Question: Model Comparison Node



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Question: Model Comparison Node

In Model Studio, when you add any node to a pipeline, the Model Comparison node is added automatically.

- a. True
- b. False

Answer: Model Comparison Node

In Model Studio, when you add any node to a pipeline, the Model Comparison node is added automatically.

a. True

☒ b. False

CA: b

The Model Comparison node is automatically added only when you add a **supervised learning** node to a pipeline.

Demo: Comparing Models within a Pipeline

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Demo: Comparing Models across Pipelines

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Score Holdout Data (Out-of-Time Testing)

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Demo: Reviewing a Project Summary Report on the Insights Tab

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Demo: Registering the Champion Model

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Demo: Exploring the Settings for Model Selection

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Question: Model Comparison Node Properties



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Question: Model Comparison Node Properties

Which of the following can you specify in the properties panel of the Model Comparison node? (Select all that apply.)

- a. the depth to use when applying a lift-based measure
- b. the cutoff to use when applying an ROC-based measure
- c. the data partition to use for selecting the champion

Answer: Model Comparison Node Properties

Which of the following can you specify in the properties panel of the Model Comparison node? (Select all that apply.)

- ☒ a. the depth to use when applying a lift-based measure
- ☒ b. the cutoff to use when applying an ROC-based measure
- ☒ c. the data partition to use for selecting the champion

CA: a,b,c

You can specify all these things in the properties panel of the Model Comparison node.

Model Selection Statistics by Target Type

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Question: Model Selection Statistics



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Question: Model Selection Statistics

When you have an interval target, which of the following fit statistics can you use to select the champion model?

- a. area under the curve (C-statistic)
- b. average square error
- c. cumulative lift
- d. misclassification

Answer: Model Selection Statistics

When you have an interval target, which of the following fit statistics can you use to select the champion model?

- a. area under the curve (C-statistic)
- ☒ b. average square error
- c. cumulative lift
- d. misclassification

CA: b

You can use the average square error for both categorical (class) and interval targets. The other fit statistics shown here are applicable only to categorical targets.

6.2 Model Deployment



Scoring and Managing the Champion Model



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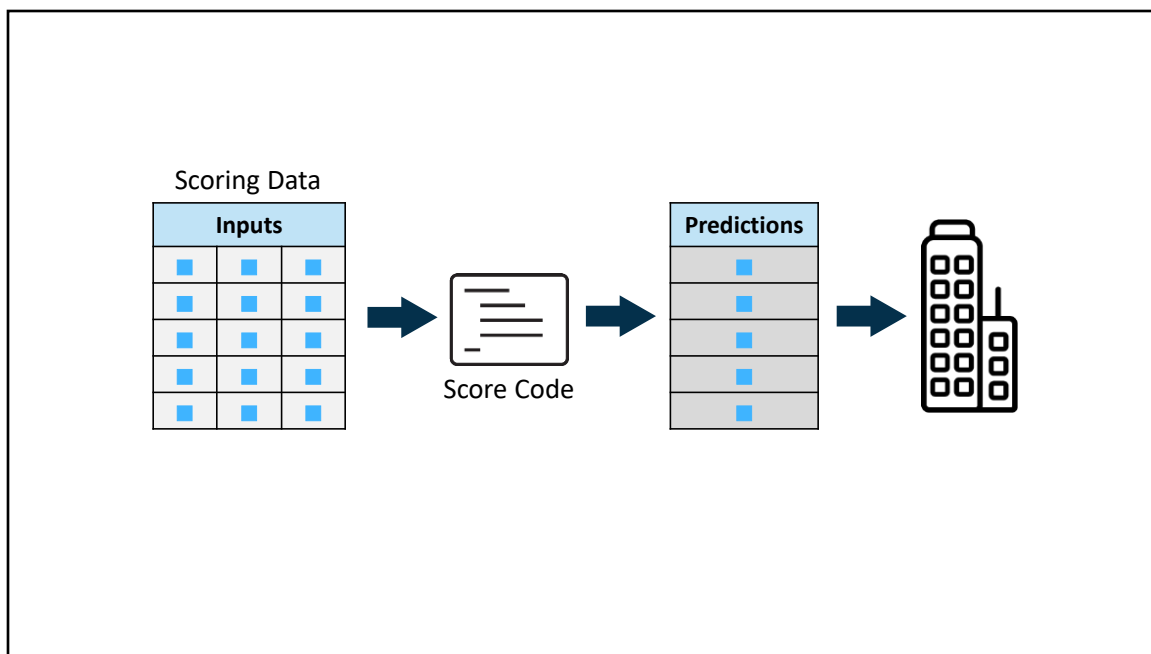
Scoring and Managing the Champion Model



Essential Deployment Tasks

- ✓ Assess models.
- ✓ Compare models.
 - Score the champion model.
 - Assess model bias.
 - Monitor model performance over time.
 - Update the model as needed.

After selecting a champion model, it's time to deploy the model -- that is, put it to use in a production environment. All tasks performed earlier in the analytics life cycle lead to this task: generating predictions through scoring.



For scoring, the model is first translated into another format (typically, score code). In the SAS Viya environment, score code is a SAS program that you can easily run on a scoring data set. The model is then applied to the scoring data set to obtain predicted outcomes. Based on the predictions from the model, the enterprise makes business decisions and takes action.

Score Code and Model Deployment

To explore this learning resource in greater detail, please visit the course page on learn.sas.com. There, you'll find comprehensive course materials, tutorials, and updates to support your learning journey.



Internally Scored Data Sets in Model Studio

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Introducing SAS Model Manager

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Demo: Viewing the Score Code and Running a Scoring Test

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SAS Model Manager Tabs

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Assessing Model Bias



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Assess Model Bias



Essential Deployment Tasks

- ✓ Assess models.
- ✓ Compare models.
- ✓ Score the champion model.
 - Assess model bias.
 - Monitor model performance over time.
 - Update the model as needed.

Fairness and Bias

Every model application has the potential to affect separate segments of the user population differently. It's important for you to understand the biases present in your models to take proper mitigating actions. Bias assessment in Model Studio comes in the form of a new tab in the Results of a supervised learning model called Fairness and Bias. To view these reports for a particular classification variable, such as Sex, you must select the "Assess this variable for bias" option in the Data tab of a Model Studio project.

See the next page and the resources mentioned therein for more details.

Assessing Model Bias and Relevant Resources

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Monitoring and Updating the Model



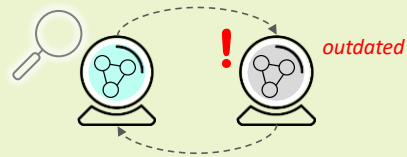
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Monitoring and Updating the Model

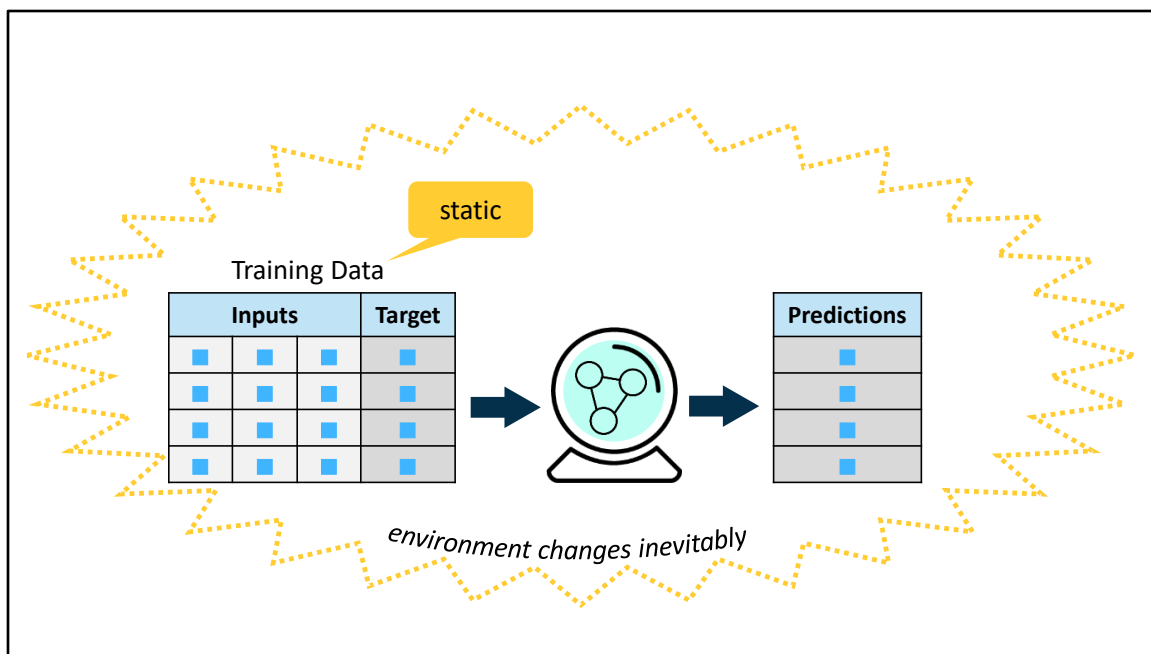


Essential Deployment Tasks

- ✓ Assess models.
- ✓ Compare models.
- ✓ Score the champion model.
- ✓ Assess model bias.
- Monitor model performance over time.
- Update the model as needed.

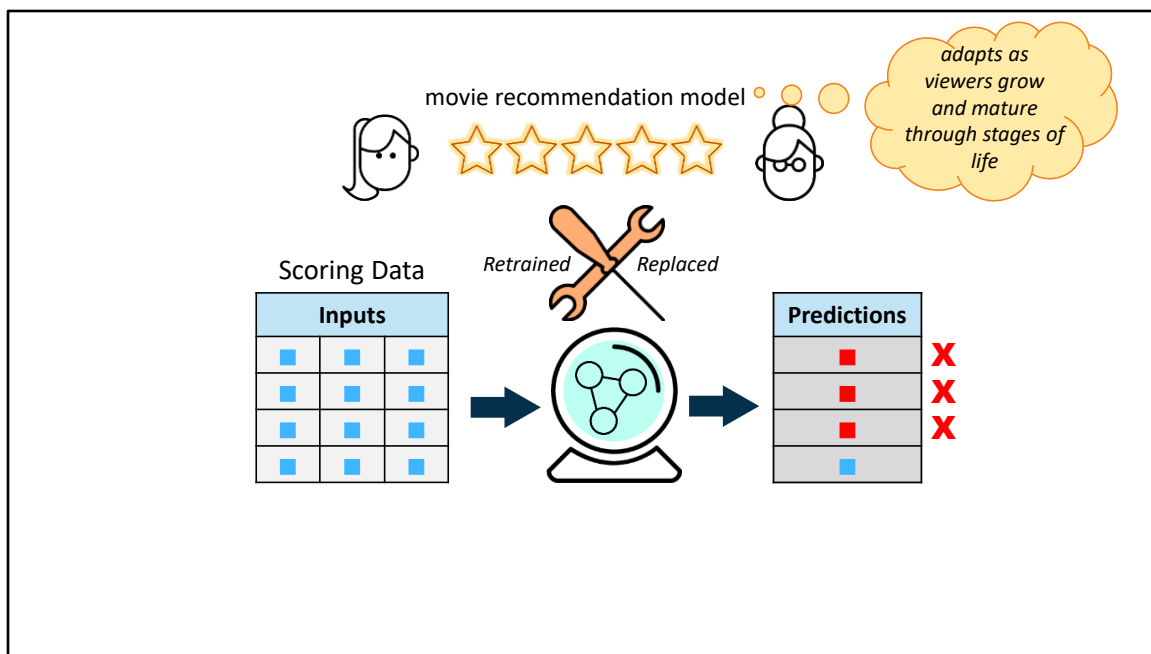


After a model is deployed, you must closely monitor its accuracy and performance over time. At some point, the model might become outdated. It is then that we would likely update the model so that it can continue to perform well. Let's discuss how this might happen.



Models are trained on static snapshots of data.

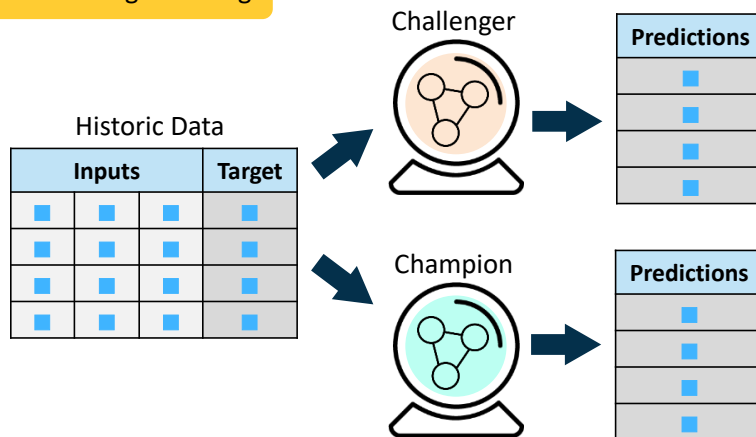
After the model is deployed, the environment inevitably changes and becomes less and less like the conditions that were captured in the training data.



As a result, the model's predictions typically become less accurate. For example, consider a movie recommendation model that must adapt as viewers grow and mature through stages of life.

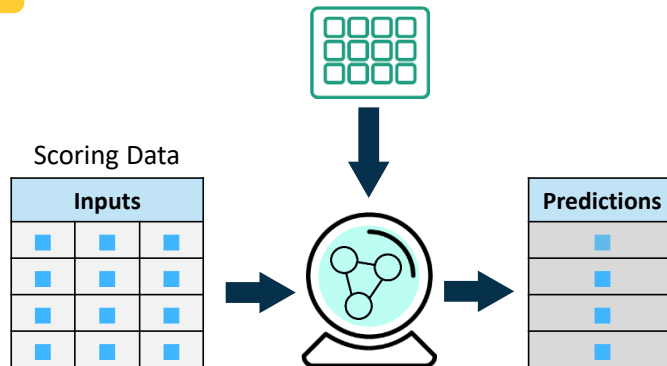
After a certain period, the error rate on new data surpasses a predefined threshold, and the model must be retrained or replaced.

Champion-Challenger Testing



Champion-challenger testing is a common model deployment practice. This method compares the performance of a new, challenger model with the performance of the deployed model on a historic data set at regular time intervals. If the challenger model outperforms the deployed model, the challenger model replaces it. The champion-challenger process is then repeated.

Online Updates



Another approach to refreshing a trained model is through online updates. Online updates continuously change the value of model parameters or rules based on the values of new, streaming data. It is prudent to assess the trustworthiness of real-time data streams before implementing an online modeling system.

Question: Scoring



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Question: Scoring

Which of the following statements about scoring are true?
(Select all that apply.)

- a. In the SAS Viya environment, score code is a SAS program.
- b. You use the validation data again for scoring.
- c. You can export score code from SAS Model Manager.

Answer: Scoring

Which of the following statements about scoring are true?
(Select all that apply.)

- ☒ a. In the SAS Viya environment, score code is a SAS program.
- ☐ b. You use the validation data again for scoring.
- ☒ c. You can export score code from SAS Model Manager.

CA: a,c

For scoring, you use scoring data (which is new data), not validation data (which was used during model building).

6.3 Lesson Conclusion



Lesson Quiz

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